(a) Gemma 4 31B's scores on failing patches

Share of failing patches or programs by score (0–100)

Mini issues, mixed harness

Own	n=80	61%
Other writers	n=103	20%

Mini issues, matched harness

Own	n=21	57%
Other writers	n=103	20%

Held-out issues

Own	n=85	61%
Other writers	n=393	35%

SWE-bench Lite, another team's run

Own	n=130	63%
Other writers	n=695	40%

LiveCodeBench programs, no agent

Own	n=124	50%
Other writers	n=403	8%

Score 0–49 50–94 95–99 100

(b) Self-judging gap

AUROC lost on own patches vs. outside judges, 95% CI

Gemma 4 31B judging its own patches

Mini, mixed harness	150 own patches	**+0.135**
Mini, matched harness	42 own patches	**+0.066**
Held-out issues	180 own patches	**+0.074**
Held-out issues	143, re-asked left out	+0.030
SWE-bench Lite	another team's run, 293 own	**+0.078**
LiveCodeBench	no agent, 250 own programs	**+0.061**
Five other models	pooled, own held-out patches	−0.001

−0.1 0 0.1 0.2

← better on own worse on own →